

A Fair Rock Game (fairgame)

Alice and Bob are playing a game: they have a pile of N rocks, and in every turn the current player takes at least 1 rock and at most K rocks. Alice makes the first move, then Bob does the second, and so on as they alternate turns.



Figure 1: Playing with rocks.

If one player takes an odd number of rocks, he has to pay M euros. When the pile is empty, the player who made the last move gets P euros and the other one gets Q euros. After the game ends, Alice will end up with an amount of euros, let's call it X , and Bob will end up with another amount of euros, let's call it Y .

They both play optimally, which means Alice wants to maximize the value $X - Y$ and Bob wants to minimize it. Find out the value of $X - Y$.

 Among the attachments of this task you may find a template file `fairgame.*` with a sample incomplete implementation.

Input

The first line contains the integers N, K, M, P, Q which describe the game.

Output

You need to write a single line with an integer: the final difference in euros $X - Y$.

Constraints

- $1 \leq K \leq N \leq 5\,000\,000$.
- $1 \leq M, P, Q \leq 5\,000\,000$.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points) Examples.

- **Subtask 2** (10 points) $K = 1$.

- **Subtask 3** (15 points) $N \leq 5$.

- **Subtask 4** (25 points) $N \leq 1000$.

- **Subtask 5** (10 points) $N \leq 100\,000, K \leq 900$.

- **Subtask 6** (20 points) $N \leq 100\,000$.

- **Subtask 7** (20 points) No additional limitations.


Examples

input	output
6 3 5 4 2	2

In the **first sample case** one possible optimal strategy is:

- Alice removes two rocks, paying nothing. Four rocks are left on the stack.
- Bob removes three rocks, paying five euros. One rock is left on the stack.
- Alice must remove the last rock, paying five euros. She gets four euros, Bob gets two.

In the end, Alice ends up with -1 euros (she lost 1 euro), while Bob finishes with -3 euros (he lost 3 euros). In total, the difference between what Alice and Bob ended up with is 2 euros.